

WHERE INDIA CAN WIN FIRST

For all the discussion around self-reliance, industry experts argue that the more practical question is not whether India can localise everything, but where it can realistically create value first.

Backend activities such as packaging, testing, modules and subsystem integration offer the most immediate opportunities because they require significantly lower ecosystem maturity than leading-edge semiconductor fabrication, says Manickam.

India also has opportunities to develop domestic capabilities in semiconductor-grade chemicals, materials and components through joint ventures, technology partnerships and strategic acquisitions.

One example is leadframes, a critical packaging material used extensively in legacy semiconductor packages. Leadframes are essentially copper structures coated with nickel-palladium plating and widely used across the OSAT facilities approved under the government's semiconductor programme. "Currently, 100% of leadframes are imported from China and Japan," says Manickam. "This could easily be made in India, although the cost may be slightly higher initially due to lower volumes."

ACHIEVING CHIP SOVEREIGNTY

Despite the scale of India's import dependence today, industry experts believe it can be reduced substantially over the next decade as domestic manufacturing capacity and supplier ecosystems mature.

According to estimates by Fab Economics, import dependence could decline from more than 85% of fab project costs and around 50% of wafer manufacturing costs today to roughly 68% and 30%, respectively, by 2030. By 2035, those figures could fall further to 55% and 18%, bringing India closer to the supply-chain depth seen in established semiconductor hubs.

Getting there, however, will require far more than a handful of fabrication plants.

"Following ISM 1.0, India requires two additional incentive packages—ISM 2.0 and ISM 3.0—by 2030, aggregating to nearly \$40 billion in subsidies across the Centre and states. This has to be followed by another two ISM 4.0 and 5.0 with incremental subsidy across the Centre and states of \$20 billion each," adds Faruqui.

According to Faruqui, the next phase of the semiconductor mission should focus on developing domestic materials, bare wafers and batch equipment manufacturing. Subsequent phases must expand

into advanced packaging, glass and panel substrates, panel-level packaging technologies and eventually semiconductor manufacturing equipment itself.

In essence, India is not building a factory. It is attempting to build an entire industrial ecosystem, layer by layer. Yet, creating domestic suppliers is only half the battle. "The consumers of these inputs, namely fabs and OSATs, will have stringent quality and reliability pre-requisites and globally accepted testing processes while using any of the core ecosystem components to ensure that their output (processed wafer or final packaged component, etc.) qualifies the stringent end-customer requirement," says Bhatnagar of KPMG.

REDEFINING SELF-RELIANCE

For all the focus on reducing import dependence, industry experts argue that India's semiconductor ambitions should not be measured by complete self-sufficiency.

ACCORDING TO NITI AAYOG'S LATEST ROAD MAP, INDIA SHOULD AIM TO MEET 15-25% OF DOMESTIC SEMICONDUCTOR DEMAND THROUGH LOCAL PRODUCTION BY 2030

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According to NITI Aayog's latest roadmap, India should aim to meet 15-25% of domestic semiconductor demand through local production by 2030, taking it up to 35-50% by 2035. Equally important is increasing value retention within the country, from 35-40% of the value of every chip consumed in India by 2030 to as much as 55-70% by 2035. Those targets are not unattainable but extremely ambitious.

According to McKinsey, roughly 60% of semiconductor materials and chemicals used by the US ecosystem continue to depend on foreign sources. "The goal is not to make everything in India, but to reduce and diversify risk for companies in India. It is about ensuring that no single supplier, customer or country can disrupt our capabilities," said Prithvideep Singh, General Manager at CDIL Semiconductors.

India may never manufacture every machine but the real prize would be to secure capability, capacity and influence to ensure that its digital future is determined as much in India as it is anywhere else. **BT**